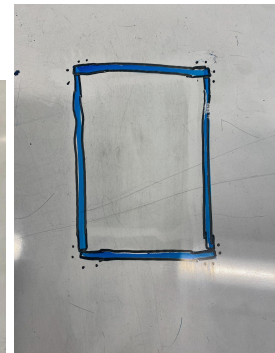
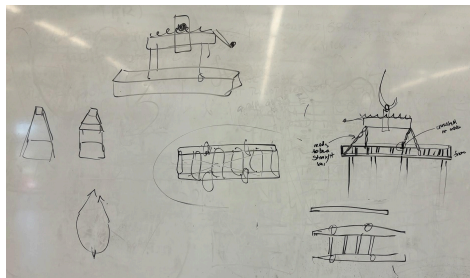
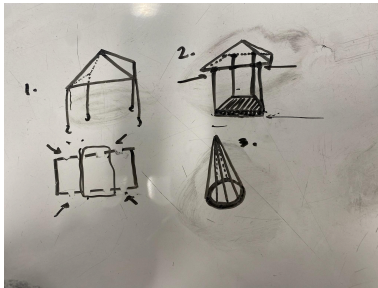


Element G

The Details of Our Construction Steps include:

1. Acquire unistrut metal for frame construction: Start by getting the metal needed for the frame.
2. Cut unistrut into 2 4ft pieces and 2 6ft pieces: Cut the metal into the required lengths to form the frame.
3. Acquire corner brace pieces to connect the unistrut into a rectangle: Get the corner braces to join the pieces together into a rectangular shape.
4. Acquire engine balance lift: Obtain the engine balance lift necessary for the project.
5. Dismantle old prototype to reuse nuts and bolts: Take apart the previous prototype to salvage usable nuts, bolts, and other components.



Our design went through multiple iterations from the original blueprint as seen above.

The physical prototype faced several issues, such as excessive torque on the metal frame, lack of available materials, and challenges in connecting the frame, balance, and straps efficiency. Each iteration aimed to address these problems, leading to improvements and adjustments in the design.

The original prototype we encountered had various problems, including excessive torque on the metal frame, a lack of durable materials, balance, and straps underlining the mat. These issues highlighted the need for better materials and a more efficient connection method.

Material Details:

1. Variety of bolts (including nuts, washers, u-hooks, screws): Used as connection pieces for the frame and other necessities. Reused from other projects, so they were free.
2. 4 26ft rolls of yellow straps: Used to hang from the frame in order to connect the frame to their already existing mat. These were reused from past prototypes and were free.
3. 30ft unistrut/superstrut: Used to make the frame. Purchased from Home Depot for \$175.
4. Carabiners: Used to hook the straps onto the existing mat. These were donated to us by Bill and were free.
5. Corner brackets: Used to attach the cut pieces of superstrut together at the corners. Purchased from Home Depot for \$45.

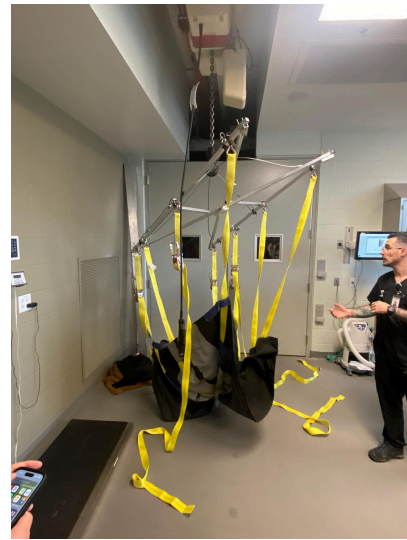
The final prototype did not include all the features we originally intended because we couldn't connect each section as efficiently as planned. These were our original design requirements; Weight-bearing capacity (50-1100 pounds): The frame needs to support a wide range of weights without failure. Balance: It should be able to balance weight 50/50 across its center of mass for a period of time. Durability: The frame should not have critical failures during use (e.g. breaking). Ensuring the device does not hurt the animals: The design should be safe and not cause additional harm to the animals. Removability: The frame should be easy to remove after use (e.g. in a approx. one to two time frame). Ease of use: Minimize the number (approx. 5, but could vary depending on human error/personal choices) of steps required from loading to

lifting. Time to load: The loading process should be quick and efficient (see removeability).

Consequently, it is unlikely to meet criteria such as the weight requirement and stability, which is unfortunately similar to the original prototype provided by last year's group. With the amount of iterations and testing attributes (seen below) there was, however small, improvement seen from the original design.



(our design – old design)



Our testing Testing Attributes include:

1. The amount of weight it can hold: This tests the weight-bearing capacity of the device by using materials such as the engine lift which has been tested by professionals and ensured it can hold at least 1500lbs.
2. The degree of the tilt: We tested the balancing capability of the frame by using levels and testing disproportionate weights to simulate the animals it will be used for.
3. Testing Evidence: Once the frame is connected to all the pieces, you can hang it from the ceiling as intended. Then start adding weights to the mat to test the balancing

mechanisms and see if it meets the weight requirement when lifted. This will demonstrate whether the frame can hold the intended amount of weight and maintain balance.

Non-Testable Attributes:

1. The look of the animal lift: This cannot be tested mathematically because it is an aesthetic attribute with no discernible measurements. It requires a subjective assessment from primary stakeholders who will be interacting/utilizing the lift in the future.

Our final prototype iteration shows how the design evolved, the materials used, the construction process, and the testing of critical attributes, while also acknowledging the limitations and non-testable aspects of the project. This demonstrates the progress we have made thus far. This is all reflected in the comments made by field experts (shown below)

Total: <u>34</u> /42	This should concern you... this is our client.		
Judge name: <u>Stephen Wancra</u>			
Notes: <u>Consider Any points of contact with animal safety. cushioning on hoist connections</u>			
<u>$\bar{x} = 34$</u>			

3 /3	neither present nor valued in the design process.	present.	present and valued in part of the design process.	present throughout process.
Total: <u>35</u> /42				
Judge name: <u>Jeremy Qualls</u>				
Notes: - would have liked if all members had a part in presentation - Great design on the leveling system - proper iteration happening with future changing of metal.				